

Symmetry-Shaped Singularities in High-Temperature Superconductor H₃S

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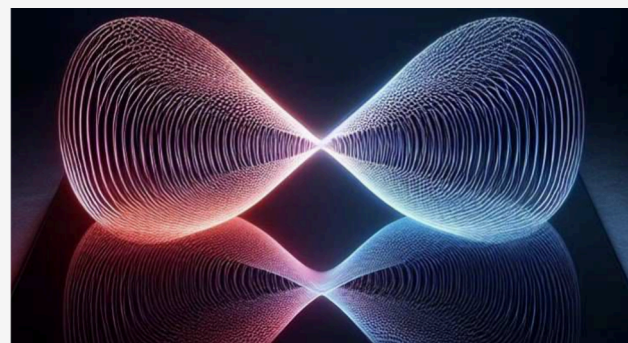


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ABSTRACT: The superconducting critical temperature of H₃S ranks among the highest measured, at 203 K. This impressive value stems from a singularity in the electronic density-of-states, induced by a flat-band region that consists of saddle points. The peak sits right at the Fermi level, so that it gives rise to a giant electron–phonon coupling constant. In this work, we show how atomic orbital interactions and space group symmetry work in concert to shape the singularity. The body-centered cubic Brillouin Zone offers a unique 2D hypersurface in reciprocal space: fully connecting squares with two different high-symmetry points at the corners, Γ and H, and a third one in the center, N. Orbital mixing leads to the collapse of fully connected 1D saddle point lines around the square centers, due to a symmetry-enforced s-p energy inversion between Γ and H. The saddle-point states are invariably nonbonding, which explains the unconventionally weak response of the superconductor's critical temperature to pressure. Although H₃S appears to be a unique case, the theory shows how it is possible to engineer flat bands and singularities in 3D lattices through symmetry considerations.



INTRODUCTION

The past decade has seen substantial scientific and public interest in hydrogen-rich superconductors with a high critical temperature (T_c).^{1–7} Light hydrogen brings high-frequency phonons, which allow for high T_c 's in conventional superconductivity. It was on this basis that Neil Ashcroft theorized metallic hydrogen (H) to be a high- T_c superconductor, a 55-year old prediction that is still propelling the field.⁸

Because the metallization of elemental H takes place at mammoth pressures—our current best prediction is 577 GPa⁹—the notion is to 'alloy' it with one or more chemical elements that promote the metallic state.^{10,11} With this idea, the pursuit for new superconductors entered the territory of chemistry. Recent advances in computational structure prediction and high-pressure synthesis have led to new hydride superconductors with unusual stoichiometries that, still, require extreme conditions to be stable. H₃S and LaH₁₀, the two materials with the highest confirmed T_c 's of respectively 203 and 250 K, were both predicted *in silico* before they were synthesized.^{2,12} Several groups have reproduced the results for H₃S since, and the synthesis and characterization methods have improved.^{13–16}

H₃S looks nothing like an alloy of high-pressure H. It consists of perfect SH₆ octahedra in a body-centered cubic (bcc) lattice, with short S–H bonds of 1.5 Å (Figure 1). This is no surprise; one S brings twice as many valence electrons to the table as three H's, so we should not expect just H to control the electronic properties. As a matter of fact, the high

T_c stems from an unusually large electron–phonon coupling constant λ of about 2, which itself has been linked to strong covalent S–H bonding.^{17,18}

Yet it has remained unclear what exactly is that connection between covalent S–H bonding and the giant λ . More precisely, we are unclear on how the bonding shapes a conspicuous spike in the density of states (DOS) at the Fermi level E_F ,^{19,20} the main factor to why λ is as big as it is. Akashi has shown this spike to originate from 1D lines of saddle points on a band hypersurface in reciprocal space.²¹ With $\mathbf{k}$ the electron wave vector, and E the energy, the points on these lines carry the property that $|\nabla E(\mathbf{k})| = 0$. In other words, they are zero-gradient points (ZGPs), and along the line, the band is almost perfectly flat. This flatness leads to a point of divergence in the electronic DOS, which is sometimes referred to as a Van Hove singularity.^{22,23} In this work, we show how an interplay between orbital interactions and space group symmetry generates lines of ZGPs in specific regions of the first Brillouin Zone (BZ). The text is organized as follows: (i) we describe the bonding in H₃S in reciprocal space (ii) we

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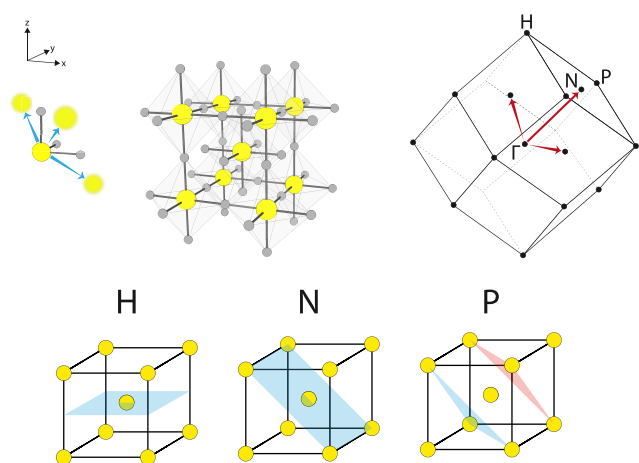


Figure 1. H_3S primitive unit cell and lattice vectors (blue) building a body-centered cubic lattice. The corresponding first Brillouin Zone is shown to the right, defined by reciprocal-lattice vectors in red. On the bottom, crystallographic planes corresponding to the (001), (101) and (111) directions, which respectively correspond to points H, N and P in the Brillouin Zone. The color of the planes indicates phase, with π in blue and 2π in red (for point P).

relate the bonding to the emergence and geometry of the ZGP lines and (iii) we discuss some general concepts and consequences.

RESULTS AND DISCUSSION

Structure in Real and Reciprocal Space. Like any bcc lattice (space group $Im\bar{3}m$, point group O_h), H_3S can be viewed as two **interpenetrated**, simple-cubic (sc) sublattices, and we define a symmetric set of vectors to span the lattice from a one-formula-unit cell. Denoting the S–H bond length by δ , the blue vectors in Figure 1 point from one sc sublattice to the other. They are

$$\begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{pmatrix} = \delta \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \quad (1)$$

Here, columns are the Cartesian coordinate system (x, y, z). We remind that bcc real space lattice vectors take the form of a face-centered cubic lattice in reciprocal space:

$$\begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{pmatrix} = \pi/\delta \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \quad (2)$$

To maintain a practical perspective, we orient vectors $\mathbf{b}_2$ and $\mathbf{b}_3$ along the negative y -direction. The red vectors shown on the right of Figure 1 are half the magnitude of $\mathbf{b}_1$, $\mathbf{b}_2$ and $\mathbf{b}_3$, so that they lie exactly within the BZ. They define three symmetry-equivalent points N with reduced coordinates $(\frac{1}{2}, 0, 0)$, $(0, \frac{1}{2}, 0)$ and $(0, 0, \frac{1}{2})$ along $\mathbf{b}_1$, $\mathbf{b}_2$ and $\mathbf{b}_3$. The phase factor associated with these points N corresponds to (101), (110) and (011) planes drawn within the conventional (two-formula) unit cell. Similarly, point H with reduced coordinates $(\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$ corresponds to a phase factor along the (001) direction, and point P, with reduced coordinates $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, to

one along the (111) direction. The bottom of Figure 1 depicts these crystallographic planes, with colors representing the phase: π in blue and 2π in red (for point P).

Crystal Orbitals. In line with the seminal work of Zak,²⁴ we define the electronic band as a set of localized crystal orbitals (COs) that are invariant under all operations of the space group. This is consistent with the general form of the one-electron Bloch function $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u(\mathbf{r})$ – where $\mathbf{r}$ is the position vector – and $u(\mathbf{r})$ a periodic basis. In our case, we employ linear combinations of atomic orbitals (LCAOs), which means the basis $u(\mathbf{r})$ is a LCAO that satisfies $u(\mathbf{r} + \mathbf{R}) = u(\mathbf{r})$ for any lattice vector $\mathbf{R}$.

The wave vector $\mathbf{k}$ sets the phase factor $e^{i\mathbf{k}\cdot\mathbf{r}}$ and with that, the local point group symmetry for $\psi_{\mathbf{k}}(\mathbf{r})$. That is because $\mathbf{k}$ itself carries a group, which is the set of point group operations g that fix it. In mathematics, it is referred to as the stabilizer subgroup, and it herein takes the form: $\text{Stab}_{O_h}(\mathbf{k}) = \{g \in O_h \mid g \cdot \mathbf{k} = \mathbf{k} \pmod{\mathbf{K}}\}$. The point group is O_h and the notation indicates that a point $\mathbf{k}$ is equivalent to $\mathbf{k} + \mathbf{K}$, with $\mathbf{K}$ a reciprocal-lattice vector. The argument shows that the number of elements, in this subgroup, is dependent on $\mathbf{k}$. For most points in the BZ, only the identity operation (E) fixes $\mathbf{k}$, and the point group classification of $\psi_{\mathbf{k}}(\mathbf{r})$ is C_1 . But at a number of high-symmetry points (HSPs), and on the lines that connect them, $\mathbf{k}$ is retained under more operations.

HSPs (with the exception of Γ) lie at the borders of the BZ; at vertices, edge or face centers. In such, they set boundary conditions for $\mathbf{k}$, and this works in more than one way. Bands tend to have their extrema at HSPs, and if there is enough symmetry, it is at these points that they tend to stick together in degeneracies that do not interact with each other.²⁵ In the case of s and p orbitals in a cubic lattice, as we have here, these degeneracies are distinct by simple parity when taken to the atomic limit. Thus, HSPs act as nodes that ‘dress up’ the BZ with bands, often in a chemically comprehensible way.

The group of $\mathbf{k}$ at Γ is the full point group of the lattice by definition. At HSPs H, N and P, the respective point groups are O_h , D_{2h} and T_d . This reveals an important symmetry property of any bcc lattice, namely that both at Γ and H, $\mathbf{k}$ is fixed by all elements of the O_h point group. What this breaks down to, as we will see in more detail, is that the phase factor associated with a shift from Γ to H does not alter bonding within SH_6 octahedra (local O_h orbital symmetry is retained). It rather changes bonding between the octahedra, and more precisely; between octahedra associated with the different sc sublattices, with a phase difference of π along the (001) plane (Figure 1). The phase factors at N and P change bonding within octahedra along two and three directions, respectively.

Let us now interpret the COs from the LCAO perspective. With four atomic orbitals for S ($s + p$), and three (s) for the Hs, we expect 7 valence bands for H_3S . These bands are shown in the top of Figure 2, in the well-studied electronic band structure of H_3S . Drawn in thinner black lines, there are additional bands from S 3d orbitals that have come down in relative energy under pressure (due to the fact they have no radial nodes).²⁶ None of these cross E_F , however, and we focus our analysis on the set of 7. The (increasing) energy ordering at Γ is A_{1g} (all-s σ), 3-fold T_{1u} (S 3p), 2-fold E_g (H 1s) and A_{1g} (all-s σ^*). The symbols denote irreducible representations (irreps). We obtain a known result from ligand field theory that will be critical to our analysis; by virtue of their parity, s

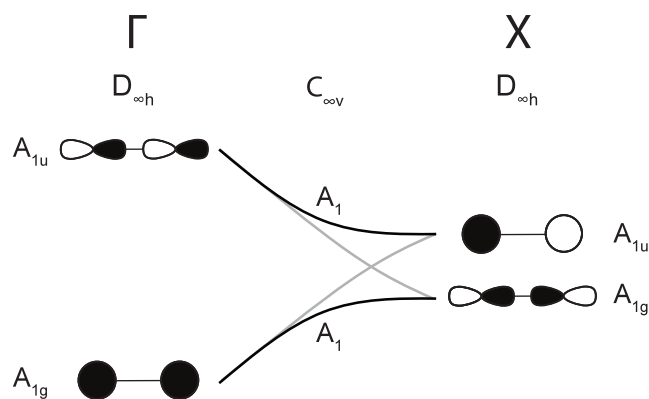


Figure 3. Schematic showing the mixing of s and p orbitals in a 1D chain, and the resulting avoided crossing. The gray lines serve as help by indicating the fictional scenario of no mixing, with a crossing as result. The black lines show the 'real' bands, with mixing. At Γ and X , the group of wave vector $\mathbf{k}$ is $D_{\infty h}$, with inversion symmetry, so that s and p_σ cannot mix. At any point in between, the group of $\mathbf{k}$ is $C_{\infty v}$ and the two can mix.

(T_{1u}) at H. And it goes vice versa for the T_{1u} band at Γ , which turns into all-s E_g at H.

If the slopes of the interacting bands have an opposite sign, ACs always lead to extrema or inflection points (saddle points) away from HSPs; in the band structure in Figure 2 these are marked by yellow discs. The associated exchanges of AO character are indicated by double-headed arrows in the diagram on the bottom of Figure 2. This shows how mixing between three totally symmetric A_1 bands of mixed (s-p) character gives rise to all ACs along the path, much like mixing between bonding, nonbonding and antibonding combinations of the same symmetry in molecules. In such, we can also consider the system as the interplay of nonbonding central A_1 levels – which we will indeed show to be largely nonbonding around E_f – with bonding and antibonding counterparts (below and above). The respective levels are colored blue in the band structure of Figure 2, with a thicker line for the central ones. The only possible challenge to this much simplified picture comes from the zigzag shape in the Γ -H segment, its origin an interaction by yet higher-lying (A_1) states. Yet this perturbation occurs more than 10 eV above E_f and shall not be significant to our analysis. The bottom all-s σ band remains too low in energy to see significant mixing with p orbitals at any point.

It is the maximum on the H–N path – at about two-third on the way and exactly at E_f – which was identified to be one ZGP on lines of saddle points that give rise to the singularity (bottom large yellow disc in the band structure of Figure 2). These lines form an almost closed loop around N.²¹ Comparing the band structure with the CO diagram below will reveal how the inflection point appears because of a second s-p energy reordering. This restores the original ordering found at Γ , although it may not immediately be obvious how. Unlike Γ and H, the smaller D_{2h} point group at N allows for S–H bonding. Two T_{1u} and two E_g bands combine to form B_{2u} and B_{3u} bonding and antibonding combinations, where p orbitals have transformed as vectors, with a p_x and a p_z AO giving rise to two 'p_{xz}' linear combinations. These levels involve S–H bonding. The H–H interactions are significant as well; they are antibonding for B_{2u} and bonding for B_{3u} along the y-axis. This has been described by others before,^{28,29} and it helps to create

a large energy gap between the occupied B_{2u} and B_{3u} levels at N. More importantly, both the bonding and antibonding B_{2u} combinations are A_1 along the H–N path, and they mix with the A_1 σ^* band. We thus recover the original picture of three interacting A_1 bands. The σ^* band descends steeply in energy, and crucially, dips below E_f at N, because the interactions between octahedra switch from antibonding to nonbonding. The band is not allowed to cross with the energy-ascending band that ends up as the antibonding B_{2u} combination, so the two mix and produce the AC that is responsible for our singularity at E_f . The mixing turns one of the all-p T_{1u} bands at H into all-s A_g at N – this is the second inversion. The gap between the two interacting bands (as a result of the AC) is giant, about 8 eV, which signals the commensurate orbital interactions to be rather strong.

It is important to note that the significant orbital interactions at play do not just describe S–H interactions. To further highlight this, consider just the T_{1u} degeneracies at Γ and H. Between the two HSPs, there is only a change in covalent S–S interactions between the sc sublattices, yet this change covers almost 10 eV. The same can be said about the energy change of the E_g orbitals at Γ and H, with only a change in (sublattice) H–H interactions.

Let us look at what happens along the rest of the path. From N back to Γ , we lose the S–H bonding again. The bonding B_{2u} and B_{3u} bands turn into all-p T_{1u} , and the antibonding combinations into all-s E_g and A_{1g} . The central levels stay totally symmetric all the way, and we again obtain three interacting A_1 bands. Although the central band shows interaction with its A_1 counterparts, it does not switch AO character along the Γ -N segment, and there is no s-p energy inversion with an associated AC along this segment. This is what we should expect; since we went through the s-p energy inversion once in Γ -H, then reinverted it in H–N, we should expect zero (or an even number) of AC-inducing inversions in a closed path back to Γ .

In moving from Γ , to P, the group of $\mathbf{k}$ goes from O_h to T_d . A C_3 axis is now retained along the path, which protects a 2-fold degeneracy. Thus, $E_g + T_{1u} + A_{1g}$ (σ^*) turn into two sets of $A_1 + E$, which then end up at P as (3-fold) T_2 bonding and antibonding H 1s/S 3p combinations. The distinct $A_1 + E$ sets separate in energy until they are 25 eV apart, and there is no s-p inversion or associated crossing. This means an inversion must appear on the path from P to H, and the AC located halfway (indicated in Figure 2) occurs, indeed, because of H 1s/S 3p orbital exchange between two A_1 bands. The two E sets interact, but do not cross. Figure S3 in the SI shows the Γ -P-H connection in a diagram, and Figure S4 shows the bands along the P–N segment. We encounter no AC on latter segment which, once more, is what we should expect; there is neither an s-p energy inversion between either Γ and N, nor is there one between Γ and P.

All the ACs near E_f derive from the s-p energy inversion between Γ and H. These are the only points in the BZ where the COs are necessarily all-s or all-p, so that they stick together in orthogonal sets that dress up the bands across the BZ. If we define them mathematically, Γ and H form the set of G-invariants. Denoting the set of wave vectors $\mathbf{k}$ in the first Brillouin Zone as $\mathcal{B}$, the G-invariants here are the elements of $\mathcal{B}$ fixed by all the operations g in the O_h point group: $\mathcal{B}^{O_h} = \{\mathbf{k} \in \mathcal{B} \mid g \cdot \mathbf{k} = \mathbf{k} \pmod{\mathbf{K}}, \text{ for all } g \in O_h\}$. This concept is generally important in mathematics, particularly in the

description of topological spaces.³⁰ As a concept in solid-state chemistry here, the set forms a pair $\{\Gamma, H\}$ from which the ACs are generated. Any path between the pair must see a ZGP. The same concept works for the earlier-discussed 1D chain of s and p orbitals discussed, and in Figure 3. For that case, the generating pair is $\mathcal{B}^{\text{D}_{\text{oh}}} = \{\Gamma, X\}$. The associated AC is known to be smoothly deformed through the energies and electro-negativities of the AOs in the corresponding limit.³¹ This suggests it may be possible to obtain control over band flatness in similar parity-enforced ACs such as those found in H_3S , i.e., a pathway to *flat band engineering*. But the Γ -H-N hypersurface on which we showed the flat-band region to appear in H_3S is 2D, not 1D – this makes things more complicated.

To facilitate a deeper understanding of why among multiple ACs only one delineates singularity-shaping lines of ZGPs, rather than mere points, we recall Akashi's analysis highlighting the significance of the extremum at N. The 2D Γ -H-N hypersurface forms fully connecting squares with Γ and H at adjacent corners, as shown in Figure 4. The fact the two HSPs

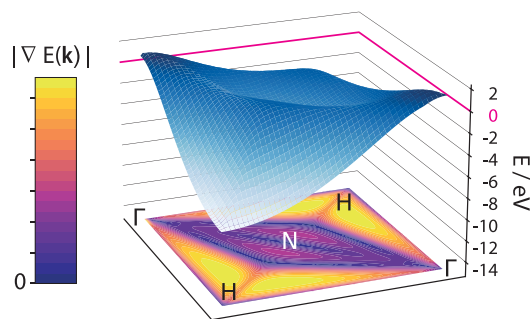


Figure 4. 2D electronic band structure showing the flat band (saddle point) region between H and N at E_f (pink). The bottom displays a contour indicating the gradient on top of the hypersurface, clearly displaying the zero-gradient lines.

have an s-p inversion between them sets up the classical saddle-point scenario for the electronic band structure, because the square will have two maxima and two minima at opposite

ends. Without an extremum at N, this would be a single saddle point. To get entire 1D lines of ZGPs instead of a single ZGP at N, we need the local extremum to “push down” the local energy level, so that it becomes a minimum. This leads to the ZGP lines to emerge around N, which can be seen in the 2D band structure plotted on top of the hypersurface in Figure 4. The figure gives a clear depiction of H_3S 's flat-band region at E_f , although the full band connectivity is a bit knottier than the plot suggests (we expand on this in the SI, section E). We can now apply our group-theoretical analysis to obtain a general understanding for similar contexts. First, we note that the extremum at N is ensured by the group of $\mathbf{k}$ at N. This is D_{2h} , a point group that contains the inversion operation. This means the extremum at N will be there for any orbital system, including a free electron dispersion. For any system with an s-p inversion between Γ and H, s and p orbitals will be split by parity at those HSPs, while setting minima and maxima at opposite ends of the hypersurface containing N. As shown in the above, this is a general orbital recipe for ZGP lines. Even though the hypersurface on which this plays out is 2D, the high 3D symmetry of the BZ presents three distinct Γ -H-N hypersurfaces. They are defined by the (001), (010) and (100) planes for H, which involve different p orbitals from the T_{1u} combinations in the s-p energy inversion. Together, the three hypersurfaces contain all permutations of the s-p energy inversion between 2-fold (E_g) and 3-fold (T_{1u}) degenerate sets. This shows that high momentum-space symmetry may not just generate flat bands, it will replicate them across multiple regions of the BZ, with a higher density of ZGP's as a result.

From the perspective of chemical bonding, the orbital analysis provides additional significant information. The electronic levels in the band at and around N can be expected to be nonbonding, since they lie in the center of a square on which the band has bonding (H) and antibonding (Γ) levels at the corners. We will now show that this is of direct physical relevance, as it explains why the T_c of H_3S shows a remarkably weak response to pressure. Let us define the quantity:

$$\frac{\partial \epsilon_{n\mathbf{k}}}{-\partial V} \quad (3)$$

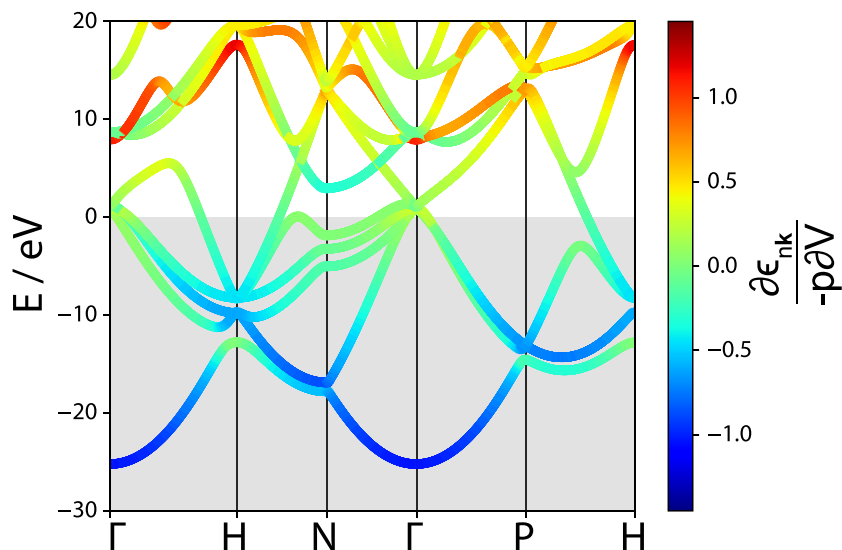


Figure 5. Dimensionless quantity to measure bonding character, plotted on top of the electronic band structure. Negative values (blue) indicate bonding, positive values (red) antibonding, and (near) zero values (green) nonbonding.

Here, $\partial\epsilon_{nk}$ denotes the change in energy eigenvalue of band index n , at point $\mathbf{k}$, upon an infinitesimal change in volume ∂V , at constant pressure p . Division by the pressure term scales the quantity, and the denominator is equal to the *work done on the system*. The dimensionless quantity is a periodic equivalent of the radial orbital-energy slope (ROS), which measures the bonding character of a molecular orbital through the response of its energy to a variation in bond length (in diatomics).³² Intuitively, a level that represents an attractive interaction, bonding, must decrease in energy when we bring nuclei closer together, and an antibonding level must go up. Figure S displays the computed quantity on top of the band structure, from bonding in blue to antibonding in red. The dimensionless quantity is well-scaled, roughly varying between -1 and 1 , and its assessment of bonding shows good agreement with the $\mathbf{k}$ -resolved Crystal Orbital Hamiltonian Populational (COHP) profile (SI section D). Importantly, regions of nonbonding are where the bands turn green; this is where the energies of the bands show effectively no response to applied pressure. Per our definition, the singularity-shaping levels are fully nonbonding. Besides the discussed zigzag, all denoted ZGPs from Figure 2 turn out to be hotspots of nonbonding states, including some far away from E_f . We reason this to be logical: for two interacting bands that shape an AC, the change in orbital character is associated with a change in bonding character, and near the point where the latter changes sign, the band slope should be zero.

CONCLUSIONS

We have shown the crucial singularity in H_3S at E_f to appear from s-p mixing, induced by an s-p energy inversion between Γ and H. This HSP pair $\{\Gamma, \text{H}\}$, mathematically defined as the set of G-invariants, is special. It consists of the only two points in the H_3S BZ where the group of $\mathbf{k}$ is O_h , and where correspondingly, all COs are necessarily all-s or all-p. These orthogonal, degenerate sets dress up the bands across the BZ, and all ACs within 10 eV from E_f appear due to the energy inversion between the two. The inversion by itself is not enough for a singularity. For that, we require the presence of a third HSP on a 2D hypersurface spun by $\{\Gamma, \text{H}\}$, which is face-center point N. The combination leads to the emergence of fully connected lines of ZGPs that bring the point of divergence in the DOS.

It is impossible to ignore some of the unique properties the bcc lattice brings to this case, and in particular the 2D square cut of the BZ (with Γ and H at corners and N in the center) on which this all plays out. Even though there is a similarity with the AC that appears in 1D s,p-systems such as cumulene, the hypersurface here is 2D, and in the simplest picture, three bands interact. These three bands can be (roughly) considered to entail interacting bonding, nonbonding and antibonding levels. The system bears resemblance to an electron-rich hypervalent system, where 'excess' electrons are seen to occupy nonbonding orbitals. Taking that analogy, one could reason that it is not surprising that E_f lies in the nonbonding regime, where we have our singularity as well. But the near-perfect alignment of the singularity with E_f is – to our current understanding – coincidental. Although this makes it hard to design an identical system, our analysis offers a new way to engineer flat bands through symmetry considerations. The energy positions of AC generating pairs like $\{\Gamma, \text{H}\}$ can (to a practical degree) be controlled through stoichiometry (and well-established trends in the periodic table). And the way we

show high momentum-space symmetry to generate the ZGP lines in H_3S suggests high-symmetry space groups should generally produce more flat bands than low-symmetry ones.

Electronic singularities have often been related to dimensional reduction in 2D systems,³³ even though their emergence in 3D systems is clearly commonplace.^{21,34} It remains to be explored how the considerations on a 2D hypersurface, as done here, connect to different classifications and models of flat-band materials and singularities.^{35–40} We note the compelling role singularities have generally played in high- T_c superconductivity. In hydrides, which are usually considered conventional superconductors, doping may align E_f more precisely with the electronic singularity, which should increase the T_c (see also the SI, section F).^{41,42} But such singularities are also observed in well-known unconventional superconductors, like the cuprates,⁴³ pnictides⁴⁴ and 'magic angle' bilayer graphene.⁴⁵ Although H_3S is believed to be a conventional superconductor, it shows unconventional traits,^{17,46} and theory predicts significant nonadiabatic effects.⁴⁷ Electronic singularities may be key to high- T_c superconductivity, and understanding how and when they emerge is important.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c02038>.

Further details on the crystal orbitals; orbitals at P and bands between P and N; bonding analysis comparison to COHP; further details on the hypersurface and the density-of-states; density of states upon doping with carbon; extensive computational details (PDF)

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Notes

The authors declare no competing financial interest.

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